

8.2 Homogeneous Linear Systems

Consider a system of linear DEs in the form :

$$\frac{d\vec{x}}{dt} = A\vec{x}(t)$$

Where A is an $n \times n$ matrix.

Suppose that A has n distinct real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ with

Corresponding eigenvectors: $\vec{k}_1, \vec{k}_2, \dots, \vec{k}_n$.

Then A is diagonalizable and can be expressed as :

$$A = PDP^{-1}$$

Where :

$$P = \left[\vec{K}_1 \mid \vec{K}_2 \mid \dots \mid \vec{K}_n \right] \quad \text{and}$$

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}.$$

$$\Rightarrow \frac{d\vec{X}}{dt} = \underbrace{PDP^{-1}}_A \vec{X}(t)$$

$$\Rightarrow \frac{d}{dt} \left(P^{-1} \vec{X}(t) \right) = D \left(P^{-1} \vec{X}(t) \right)$$

$$\text{Let } \vec{Y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{bmatrix} = P^{-1} \vec{X}(t)$$

$$\Rightarrow \frac{d}{dt} \vec{Y}(t) = D \vec{Y}(t)$$

$$\Rightarrow \frac{d}{dt} \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{bmatrix}$$

$$\Rightarrow \begin{cases} \frac{dy_1}{dt} = \lambda_1 y_1(t) \\ \frac{dy_2}{dt} = \lambda_2 y_2(t) \\ \vdots \\ \frac{dy_n}{dt} = \lambda_n y_n(t) \end{cases}$$

We now have n separable DEs that we can solve.

$$\begin{cases} y_1 = c_1 e^{\lambda_1 t} \\ y_2 = c_2 e^{\lambda_2 t} \\ \vdots \\ y_n = c_n e^{\lambda_n t} \end{cases}$$

$$\vec{Y}(t) = \begin{bmatrix} c_1 e^{\lambda_1 t} \\ c_2 e^{\lambda_2 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{bmatrix}$$

Since $\vec{Y}(t) = P^{-1} \vec{X}(t)$

then $\vec{X}(t) = P \vec{Y}(t)$

$$\Rightarrow \vec{X}(t) = \left[\vec{K}_1 \mid \vec{K}_2 \mid \dots \mid \vec{K}_n \right] \begin{bmatrix} c_1 e^{\lambda_1 t} \\ c_2 e^{\lambda_2 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{bmatrix}$$

Then:

$$\vec{X}(t) = c_1 \vec{K}_1 e^{\lambda_1 t} + c_2 \vec{K}_2 e^{\lambda_2 t} + \dots + c_n \vec{K}_n e^{\lambda_n t}$$

is the general solution.

ex: ① Solve

$$\begin{cases} x'(t) = x(t) + y(t) \\ y'(t) = 4x(t) + y(t) \end{cases}$$

with initial conditions: $x(0) = 5$, $y(0) = -2$.

Let $\vec{X}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ and

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}.$$

The system has the form:

$$\frac{d}{dt} \vec{X}(t) = A \vec{X}(t)$$

with $\vec{X}(0) = \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$

Let's find the eigenvalues of A .

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(1-\lambda)^2 - 4 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

There are two distinct real eigenvalues

$$\lambda_1 = 3 \text{ and } \lambda_2 = -1.$$

Let's now find one eigenvector for each eigenvalue.

$$\boxed{\text{For } \lambda_1 = 3}$$

$$[A - \lambda_1 I \mid \vec{0}] = \begin{bmatrix} -2 & 1 & 0 \\ 4 & -2 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x - \frac{1}{2}y = 0 \quad y \text{ is a free variable}$$

$$\begin{cases} x = \frac{1}{2}t \\ y = t \end{cases} \quad \vec{K} = t \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

↓
an eigenvector for any $t \neq 0$

$$\text{Pick } t=2 \rightarrow \vec{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\boxed{\text{For } \lambda_2 = -1}$$

$$[A - \lambda_2 I \mid \vec{0}] = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x + \frac{1}{2}y = 0 \quad y \text{ is free}$$

$$\begin{cases} x = -\frac{1}{2}t \\ y = t \end{cases}$$

$$\text{Pick } t=2 \rightarrow \vec{K}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Eigenvalue	Eigenvector
$\lambda_1 = 3$	$\vec{k}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
$\lambda_2 = -1$	$\vec{k}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

The general solution of the system is then :

$$\vec{X}(t) = c_1 \vec{k}_1 e^{\lambda_1 t} + c_2 \vec{k}_2 e^{\lambda_2 t}$$

$$\vec{X}(t) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-t}$$

Let's find c_1 and c_2 from $\vec{X}(0)$.

$$\vec{X}(0) = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & -1 & 5 \\ 2 & 2 & -2 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \end{array} \right]$$

$$\Rightarrow c_1 = 2 \text{ and } c_2 = -3$$

The solution of the IVP system is then

$$\vec{X}(t) = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t} - 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-t}$$

$$\begin{cases} x(t) = 2e^{3t} + 3e^{-t} \\ y(t) = 4e^{3t} - 6e^{-t} \end{cases}$$

$$\vec{X}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

Repeated Eigenvalues

Consider a system

$$\frac{d}{dt} \vec{X}(t) = A \vec{X}(t)$$

where A is a 2×2 matrix.

Assume that A has only one eigenvalue λ (algebraic multiplicity 2).

We will consider two cases for the eigenvectors.

Case ① If there is two linearly independent eigenvectors $\vec{k}_1$ and $\vec{k}_2$, then:

$$\vec{X}(t) = c_1 \vec{k}_1 e^{\lambda t} + c_2 \vec{k}_2 e^{\lambda t}$$

is the general solution.

Case ② If there is only one independent eigenvector $\vec{k}$ associated to λ ,

then:

$$\vec{X}(t) = c_1 \vec{k} e^{\lambda t} + c_2 (\vec{k}t + \vec{p}) e^{\lambda t}$$

is the general solution where $\vec{p}$ is a vector that satisfies:

$$(A - \lambda I) \vec{p} = \vec{k}.$$

ex: ② Solve

$$\begin{cases} x'(t) = -x(t) + 3y(t) \\ y'(t) = -3x(t) + 5y(t) \end{cases}$$

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 3 \\ -3 & 5 \end{bmatrix}}_A \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

Eigenvalues $A - \lambda I = \begin{bmatrix} -1-\lambda & 3 \\ -3 & 5-\lambda \end{bmatrix}$

$$\det(A - \lambda I) = 0$$

$$(-1-\lambda)(5-\lambda) + 9 = 0$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)^2 = 0 \rightarrow \boxed{\lambda = 2} \leftarrow \text{only one eigenvalue}$$

Eigenvectors

$$\lambda = 2$$

$$[A - \lambda I | \vec{0}] = \begin{bmatrix} -3 & 3 & | & 0 \\ -3 & 3 & | & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & -1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$x - y = 0 \quad y \text{ is free}$$

$$\begin{cases} x = t \\ y = t \end{cases} \Rightarrow \vec{K} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ are the eigenvectors for } t=1.$$

We only have one independent eigenvector

$$\vec{K} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Let's find a vector $\vec{P}$ so that:

$$(A - \lambda I) \vec{P} = \vec{K}$$

$$[A - \lambda I | \vec{K}] = \begin{bmatrix} -3 & 3 & | & 1 \\ -3 & 3 & | & 1 \end{bmatrix}$$

$$\xrightarrow{\text{RREF}} \begin{bmatrix} 1 & -1 & | & -1/3 \\ 0 & 0 & | & 0 \end{bmatrix}$$

That column
is our $\vec{P}$.

$$\vec{p} = \begin{bmatrix} -1/3 \\ 0 \end{bmatrix} \text{ satisfies } (A - \lambda I) \vec{p} = \vec{k}$$

(check it.)

The general solution is

$$\vec{x}(t) = c_1 \vec{k} e^{at} + c_2 (\vec{k}t + \vec{p}) e^{\lambda t}$$

$$\vec{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{at} + c_2 \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} t + \begin{bmatrix} -1/3 \\ 0 \end{bmatrix} \right) e^{at}$$

$$\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

$$\Rightarrow \begin{cases} x(t) = c_1 e^{at} + c_2 (t - 1/3) e^{at} \\ y(t) = c_1 e^{at} + c_2 t e^{at} \end{cases}$$

Complex Eigenvalues

Consider a 2×2 matrix A with two complex conjugates eigenvalues and eigenvectors.

Eigenvalue

$$\lambda_1 = \alpha + \beta i$$

$$\lambda_2 = \alpha - \beta i$$

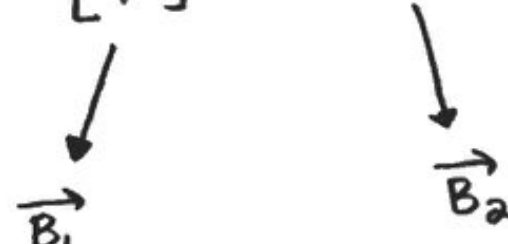
Eigenvectors

$$\vec{K}_1 = \vec{B}_1 + i\vec{B}_2$$

$$\vec{K}_2 = \vec{B}_1 - i\vec{B}_2$$

Note: If $\vec{K}_1 = \begin{bmatrix} 3+5i \\ 4+i \end{bmatrix}$, then

$$\vec{K}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + i \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$



$\vec{B}_1$ $\vec{B}_2$

We can show that:

$$\vec{X}_1(t) = \operatorname{Re} \left(\vec{k}_1 e^{\lambda_1 t} \right)$$

and

$$\vec{X}_2(t) = \operatorname{Im} \left(\vec{k}_1 e^{\lambda_1 t} \right)$$

are two linearly independent solutions

of $\frac{d\vec{X}}{dt} = A \vec{X}(t)$ so that

$$\vec{X}(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t)$$

is the general solution.

$$\vec{k}_1 e^{\lambda_1 t} = (\vec{B}_1 + i \vec{B}_2) e^{(\alpha + i\beta)t}$$

$$= e^{\alpha t} e^{i\beta t} (\vec{B}_1 + i \vec{B}_2)$$

$$= e^{\alpha t} \underbrace{(\cos \beta t + i \sin \beta t)}_{\text{Euler's Formula}} (\vec{B}_1 + i \vec{B}_2)$$

Euler's Formula

$$= e^{\alpha t} (\vec{B}_1 \cos \beta t - \vec{B}_2 \sin \beta t) + i e^{\alpha t} (\vec{B}_1 \sin \beta t + \vec{B}_2 \cos \beta t)$$

$$\vec{X}_1(t) = \text{Re}(\vec{k}_1 e^{\lambda_1 t}) \leftarrow \text{Real part}$$

$$= e^{\alpha t} (\vec{B}_1 \cos \beta t - \vec{B}_2 \sin \beta t)$$

$$\vec{X}_2(t) = \text{Im}(\vec{k}_1 e^{\lambda_1 t}) \leftarrow \text{Imaginary part}$$

$$= e^{\alpha t} (\vec{B}_1 \sin \beta t + \vec{B}_2 \cos \beta t)$$

$$\boxed{\vec{X}(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t)}$$

ex: ③ Solve the system

$$\begin{cases} x'(t) = -2x(t) + 6y(t) \\ y'(t) = -3x(t) + 4y(t) \end{cases}$$

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \underbrace{\begin{bmatrix} -2 & 6 \\ -3 & 4 \end{bmatrix}}_A \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

Eigenvalues : $A - \lambda I = \begin{bmatrix} -2-\lambda & 6 \\ -3 & 4-\lambda \end{bmatrix}$

$$\det(A - \lambda I) = 0$$

$$(-2-\lambda)(4-\lambda) + 18 = 0$$

$$\lambda^2 - 2\lambda + 10 = 0$$

$$\lambda = 1 \pm 3i$$

Complex Eigenvalues

$$\boxed{\begin{array}{l} \lambda_1 = 1 + 3i \\ \lambda_2 = 1 - 3i \end{array}}$$

Let's find an eigenvector $\vec{K}_1$

corresponding to $\lambda_1 = 1+3i$

$$\left[A - \lambda_1 I \mid \vec{0} \right] = \left[\begin{array}{cc|c} -2-(1+3i) & 6 & 0 \\ -3 & 4-(1+3i) & 0 \end{array} \right]$$

$$= \left[\begin{array}{cc|c} -3-3i & 6 & 0 \\ -3 & 3-3i & 0 \end{array} \right]$$

$$\xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & -1+i & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x + (-1+i)y = 0 \quad y \text{ is free}$$

$$\begin{cases} x = (1-i)t \\ y = t \end{cases}$$

Pick $t=1$ to get

$$\vec{K}_1 = \begin{bmatrix} 1-i \\ 1 \end{bmatrix}.$$

$$\lambda_1 = 1 + 3i \Rightarrow \alpha = 1, \beta = 3$$

$$\vec{K}_1 = \begin{bmatrix} 1 & -i \\ 1 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\vec{B}_1} + i \underbrace{\begin{bmatrix} -1 \\ 0 \end{bmatrix}}_{\vec{B}_2}$$

$$\vec{X}_1(t) = e^{\alpha t} \left(\vec{B}_1 \cos \beta t - \vec{B}_2 \sin \beta t \right)$$

$$\vec{X}_2(t) = e^{\alpha t} \left(\vec{B}_1 \sin \beta t + \vec{B}_2 \cos \beta t \right)$$

$$\vec{X}_1(t) = e^t \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cos(3t) - \begin{bmatrix} -1 \\ 0 \end{bmatrix} \sin(3t) \right)$$

$$\vec{X}_2(t) = e^t \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \sin(3t) + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \cos(3t) \right)$$

$$\vec{X}_1(t) = e^t \begin{bmatrix} \cos(3t) + \sin(3t) \\ \sin(3t) \end{bmatrix}$$

$$\vec{X}_2(t) = e^t \begin{bmatrix} \sin(3t) - \cos(3t) \\ \sin(3t) \end{bmatrix}$$

The general solution is

$$\vec{X}(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t).$$
